LEENA RAJAN KATKAR

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SUMMARY

Graduate researcher at USC Viterbi specializing in large language models, generative AI, and neuromorphic computing. Proven track record building and validating end-to-end ML pipelines (PyTorch, JAX) and scalable backend systems (RESTful APIs, MySQL, PHP, Java). Experienced across the full software lifecycle — from research experimentation to production-grade deployment — with a strong foundation in data structures, algorithms, and distributed systems. Seeking Apple's Software Engineering Internship to apply deep AI/ML expertise to products used by billions.

EDUCATION

M.S., Computer Science · *University of Southern California — Viterbi School of Engineering* Aug 2025 – May 2027 | Los Angeles, CA

- Coursework: Artificial Intelligence · Machine Learning · Deep Learning · Distributed Systems

B.E., Information Technology · *Saraswati College of Engineering* 2021 – 2025 | India

- GPA: 3.7 / 4.0 · Coursework: Advanced Data Structures & Algorithms · Computer Network Security · Big Data Analytics · OS · Database Management · Software Engineering

EXPERIENCE

Research Assistant — Generative AI & LLMs · *University of Southern California* Nov 2025 – Present

- Designed and executed quantitative PyTorch experiments benchmarking LLM architectures (transformer-based) for generative AI tasks, applying statistical metrics for performance evaluation.
- Implemented NLP pipelines for model output benchmarking; applied probability theory and error analysis to validate generative model efficiency at scale.
- Contributed to efficient AI systems research by integrating model validation algorithms and producing reproducible, data-driven findings.

Research Intern — AI & Robotics · *DRDO (Defence Research and Development Organisation)* Jan 2025 – Mar 2025

- Built end-to-end ML pipelines in JAX integrating multi-sensor data streams and video/image processing (OpenCV) using regression, classification, and time-series analysis.
- Designed and validated defense-grade AI pipelines; applied statistical validation to improve model accuracy and reliability under real-world conditions.

Project Developer — ML Systems · *Ural Federal University* Sep 2024 – Jul 2024 | Russia

- Developed and evaluated a GAN-based recommendation system at scale (10,000+ users) using deep learning, validating via iterative model experimentation.
- Built scalable data pipelines in Java; applied statistical analysis to improve recommendation precision through systematic A/B-style iteration.

Full-Stack Web Developer · *990 Productions* Feb 2024 – Sep 2024

- Engineered a full-stack CRM using PHP, MySQL, and RESTful APIs, achieving up to 90% efficiency gains in financial operations.
- Designed normalized MySQL schemas for scalable data management; built responsive frontend with HTML/CSS and Bootstrap.

Full-Stack Web Developer · *Rudraksha Welfare Foundation* Jun 2023 – Sep 2024

- Led a 6-member Agile team delivering 6+ live full-stack projects (HTML/CSS, Tailwind, JavaScript, PHP); improved workflow efficiency by 75% via Git-based version control.

TECHNICAL SKILLS

Languages: Python · C++ · Java · JavaScript · PHP · SQL

ML / AI: PyTorch · JAX · Scikit-learn · Hugging Face · LangChain · XGBoost · OpenCV · Generative AI · LLMs · GANs · Transformers

Backend & APIs: RESTful APIs · Node.js · PHP · MySQL · WebSockets

Frontend: HTML5 · CSS3 · JavaScript · Tailwind · Bootstrap

Data Engineering: NumPy · Pandas · Time-series · Statistical Modeling · Data Validation · Distributed ML
Tools & Practices: Git · Agile/Scrum · Model Evaluation · Reproducible ML Reporting

KEY PROJECTS

Terrain Mapping System

- Built 2D terrain segmentation using U-Net and 3D point-cloud modeling with KPConv in PyTorch; processed aerial imagery with OpenCV for accurate terrain classification applicable to autonomous navigation and defense intelligence.

Internship Provider Platform

- Designed and implemented RESTful APIs in PHP and JavaScript for a skill-based internship matching platform with real-time student–faculty collaboration via WebSockets.

Neuromorphic SNN for Dyslexia Detection

- Architected a Spiking Neural Network achieving 94.3% dyslexia detection accuracy with 85% GPU power reduction and sub-100ms response latency — published research.

PUBLICATIONS & PATENTS

- Patent: Jeevani — AI/ML Autonomous Water Rescue Device (95% efficiency, 99% rescue rate; 1st Prize, International Engineering Conference)
- GAN-Based Furniture Recommendation System — Springer (combines GAN recommendations, 3D AR visualization, multilingual support)
- Foundations of Machine Learning: Models, Methods, and Evaluation — Accepted, Awaiting Publication
- Neuromorphic Computing for Real-Time Dyslexia Detection — Accepted, Awaiting Publication
- Explainable AI Financial Fraud Detection (SMOTE + ensemble + SHAP) — Accepted, Awaiting Publication
- Edge-Based Confidence-Aware Continual Learning Framework for AI Wearables — Accepted, Awaiting Publication

HONORS & AWARDS

- Best Research Paper & Best Poster — NIELIT International Conference 2025
- Best Student Project Award 2024 — Computer Society of India
- Nirmaan 2023 Hackathon Winner
- International Jury Member — i-GEN 2026, Malaysia (Kolej Poly-Tech MARA)